P V SAIRAM SAKETH

Computer Vision Engineer | Edge AI | Deep Learning | UAV & Defense Systems

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PROFESSIONAL SUMMARY

Computer Vision Engineer with 1+ year of experience designing and deploying real-time, production-grade AI systems for UAV, defense and industrial-safety applications. Currently engineering low-latency computer vision pipelines on **NVIDIA Jetson Orin Nano** for the **CALM (Canister Launched Anti-Armoured Loiter Munition)** program while delivering **4 concurrent commercial AI projects**. Expertise in **object detection (YOLO), model optimization (PyTorch to ONNX to TensorRT FP16), DeepStream and GStreamer video pipelines, multi-object tracking, face recognition** and **Edge AI deployment**. Published researcher with a B.Tech in Computer Science Engineering (**CGPA 9.33/10**). Seeking Computer Vision, Deep Learning and Edge AI Engineer roles.

TECHNICAL SKILLS

Computer Vision & Deep Learning: Computer Vision, Object Detection, YOLOv8, YOLO26, Object Tracking (ByteTrack), Video Analytics, Image Processing, Face Detection, Face Recognition (SCRFD, ArcFace), PyTorch, Keras, ANN, CNN, OpenCV, Finite State Machine (FSM), VLMs, LangChain (basics)

Edge AI & Deployment: NVIDIA Jetson Orin Nano, TensorRT, ONNX, FP16 Optimization, NVIDIA DeepStream SDK, GStreamer, Real-Time Inference, Video Streaming, Docker, CI/CD (GitHub Actions)

Machine Learning & Data: Scikit-learn, Regression, Classification, Random Forest, SVM, KNN, Model Evaluation, Hyperparameter Tuning, NumPy, Pandas, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), SQL

Programming Languages: Python, C++, Rust, Go, SQL

Tools & Platforms: Ubuntu 22.04, Linux, Git, GitHub, GitLab

KEY ACHIEVEMENTS

- Owned **4 concurrent commercial computer vision projects** (PPE detection, X-ray Fuze defect detection, process-deviation monitoring, face-recognition head count) at Solar Defence and Aerospace Limited.
- Trained **3 YOLO26x detection models on 10,000 images**, achieving **70% mAP50 and 43% mAP50-95** for plant process-deviation monitoring.
- Converted PyTorch models to **ONNX and TensorRT FP16 engines** to cut inference latency and raise FPS for real-time UAV target detection on Jetson Orin Nano.
- Delivered **100% core-functionality test coverage** for the Zetaleap product launch; **published peer-reviewed research** on vehicle insurance fraud detection; earned a **Letter of Recommendation** as a Machine Learning Intern.

PROFESSIONAL EXPERIENCE

Executive – Computer Vision Engineer | Solar Defence and Aerospace Limited

Jul 2025 – Present | Nagpur, India | On-site | UAV Division

1. CALM (Canister Launched Anti-Armoured Loiter Munition) Project – developing real-time, AI-powered computer vision solutions for UAV-based defense applications.

- Architected and deployed object detection, recognition and identification software for UAV surveillance systems, improving target-identification reliability under field conditions.
- Developing an infrared (IR) thermal object detection pipeline for accurate tank/armor detection, extending mission capability to low-visibility and night-time conditions.
- Optimized deep learning models by converting PyTorch (.pt) models to ONNX and TensorRT FP16 engines, cutting inference latency and increasing FPS for real-time onboard processing.
- Built end-to-end DeepStream and GStreamer pipelines covering video ingestion, preprocessing, inference, post-processing, encoding and live streaming to the Ground Control Station (GCS).
- Deployed production-ready edge AI applications on NVIDIA Jetson Orin Nano, enabling low-latency embedded vision for time-critical defense operations.
- Partnered with cross-functional engineering teams to integrate computer vision models into UAV platforms, accelerating feature delivery for live defense programs.

Technologies: *Python, PyTorch, OpenCV, YOLOv8, TensorRT, ONNX, DeepStream SDK, GStreamer, NVIDIA Jetson Orin Nano, Linux, Ubuntu, Computer Vision, Deep Learning, Edge AI, Video Analytics*

2. Commercial Domain – 4 Computer Vision Projects

- **PPE Detection:** trained a model that detects whether personnel are wearing safety kit while mixing compositions at the Chakhdoh plant, supporting workplace-safety compliance.
- **PDM 557 Fuze – X-ray Defect Detection:** identifies defects in a component of X-ray Fuze images using filters, thresholding and area analysis. Collected data from the PYRO plant of Solar Defence and Aerospace Limited, annotated it, trained a **YOLO26x** model (40 images across 50 Fuze images, still fine-tuning) and built the application with **Rust and React JS** for deployment on the plant server.
- **Process Deviation of Delay Plant:** collected and annotated datasets from the Solar Industries defense plant and trained **3 YOLO26x models** (PPE, table, door open/close and fan on/off) on **10,000 images**, achieving **mAP50 of 70%** and **mAP50-95 of 43%** (batch size 16, image size 704, optimizer auto). Implemented an **FSM** to flag violations, built a **React** frontend with a **FastAPI** backend, and used **CIELAB** colour identification to trigger a process event (add IPA solution when the mixture turns light brown, until dark brown).

- **Face Recognition for Head Count:** built a CCTV-based system that recognises people and restricts actions to authorised persons. Benchmarked face detectors (YuNet, RetinaFace, SCRFD) and recognisers (SFace, ArcFace, AdaFace); **SCRFD + ArcFace** performed best. Solved the duplicate-embedding problem (two people in frame producing multiple embeddings per person) with an **ROI-based approach**: person detection first, click/touch selection and naming, then sampling at 5 FPS with configurable stop times of 10/20/30/40 s, embeddings stored in .npy format, and face recognition applied only to the top 50% of the person box for higher accuracy.

MTS Intern | Tessell

6 months | Remote

- Extracted machine types and availability zones from GCP through its API using the **Go** programming language.
- Implemented JSON validation schemas and used them to validate the entire JSON flow format on the backend.
- Performed error handling across the backend codebase and worked on workflows and cloud logs to pinpoint where the code was breaking.

Data & Python Intern | Rise 11 Technologies

4 months | Remote

- Designed test cases achieving **100% core-functionality coverage** for the Zetaleap project, contributing to a seamless product launch and positive feedback from early adopters.
- Applied Python and data analytics techniques to support product testing and content development.

Web Developer Trainer Intern | Coding Raja Technologies

3 months | Remote

- Delivered training sessions on Pandas and SQL to student cohorts, improving learners' hands-on data-analysis proficiency.
- Guided students through Exploratory Data Analysis (EDA) projects and mentored them on real-world data workflows.

Machine Learning Intern | Prodigy Infotech

3 months | Remote

- Built multiple supervised ML models for prediction tasks: House Price Prediction (Linear Regression), Dogs vs. Cats Image Classification (SVM) and Customer Purchase Prediction (KNN).
- Received a **Letter of Recommendation** for outstanding performance and technical contribution.

PROJECTS

VisionEye – Real-Time Foot-Traffic Analytics & Computer Vision Platform | Personal Project

Live: pvsairamsaketh.in

- Engineered a real-time people detection and tracking system using **YOLO26n ONNX, ByteTrack and OpenCV**, enabling IN/OUT foot-traffic counting, occupancy monitoring and 2D bird's-eye perspective mapping.
- Developed a full-stack analytics platform with **React 18, Django REST Framework, Django Channels and WebSockets**, featuring live MJPEG video streaming, interactive calibration overlays and privacy-preserving video masking.
- Automated CI/CD and cloud deployment using **GitHub Actions, Docker, Microsoft Azure and Nginx** with SSL/TLS; configured cross-platform desktop releases for macOS, Windows and Linux.

Optimized ML Model Selection for Heart Disease Classification | Independent Project

Python, Scikit-learn, Matplotlib, Seaborn

- Cleaned and preprocessed clinical data (missing values, duplicates, Label Encoding) and performed EDA using Matplotlib/Seaborn to surface actionable health-risk insights.
- Benchmarked multiple ML algorithms in Scikit-learn; identified **Random Forest Classifier** as the top performer for heart disease prediction.

Bank Loan Payment Status Prediction | Group / Minor Project

Python, Scikit-learn, Data Analysis

- Performed data cleaning, preprocessing, analysis and visualization on banking datasets to prepare high-quality model inputs.
- Identified Random Forest Classifier as the best-performing model; benchmarked against FPG-Vertical, which better captured complex repayment patterns.

Vehicle Insurance Fraud Detection using Bagging Classifier vs. MLP | Major Group Project – Published Research

International Conference on Computational Intelligence

- Built and compared a Bagging Classifier against an MLP model for detecting fraudulent vehicle insurance claims, identifying the **Bagging Classifier** as the superior approach.
- Published findings at the **International Conference on Computational Intelligence** (Dept. of IT & CSBS, K. Ramakrishnan College of Engineering), contributing a scalable approach to reducing fraud-related financial losses for insurers.

EDUCATION

B.Tech, Computer Science Engineering | SRM University, Chennai | 2021 – 2025 | **CGPA: 9.33 / 10**

12th Grade (Higher Secondary) | Taywade College, Nagpur | 2020 – 2021 | **Percentage: 98.17%**

10th Grade (Secondary) | Bhawans Ashti, Nagpur | 2018 – 2019 | **Percentage: 93.8%**

CERTIFICATIONS

- Exploratory Data Analysis with Python – Great Learning
- Python for Computer Vision with OpenCV – Udemy